Minkyu Jeon

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Research Summary

Ph.D. candidate in Computer Science at Princeton University, advised by Prof. Ellen D. Zhong. My research develops generative and representation learning methods for scientific inverse problems, with a focus on heterogeneous cryo-EM reconstruction and AI-driven de novo protein design.

Interests: AI agents for scientific discovery; de novo protein design; protein generative models; diffusion and flow matching in 3D structure spaces; cryo-EM heterogeneity; representation learning; 3D vision.

Education

Princeton University *2023–present*
Ph.D. Candidate, Computer Science — Princeton, NJ

Korea University *2020–2023*
M.S., Computer Science and Engineering — Seoul, Korea

Konkuk University *2014–2020*
B.S., Applied Statistics and Computer Engineering, double major — Seoul, Korea

Selected Research Projects

Agentic Inference-Time Scaling for De Novo Protein Design *2026–present*
Princeton University

- Building AI-agent workflows for protein binder and scaffold design that combine LLM-guided search, structural validation, and iterative sequence/structure refinement.
- Developing evaluation pipelines for hard de novo design targets, including diagnostics for structure confidence, interface quality, diversity, and reproducibility.
- Exploring how inference-time search, self-refinement, and targeted experimental routing can improve practical protein design beyond one-shot generative sampling.

Heterogeneous Cryo-EM Reconstruction *2023–present*
Princeton University

- Developing amortized and transformer-based methods for reconstructing continuous conformational heterogeneity from cryo-EM particle images.
- Led CryoBench, a benchmark suite for diverse and challenging cryo-EM heterogeneity problems, accepted as a NeurIPS 2024 Spotlight.

Publications

* denotes equal contribution.

CVPR 2026 — **CryoHype: Reconstructing a thousand cryo-EM structures with transformer-based hypernetworks**
Jeffrey Gu*, **Minkyu Jeon***, Ambri Ma, Serena Yeung-Levy, Ellen D. Zhong
paper

arXiv 2026 — **ConforNets: Latents-Based Conformational Control in OpenFold3**
Minji Lee, Colin Kalicki, **Minkyu Jeon**, Aymen Qabel, Alisia Fadini, Mohammed AlQuraishi
paper

NeurIPS MLSB 2025 — **Separating signal from noise: a self-distillation approach for amortized heterogeneous cryo-EM reconstruction**
Minkyu Jeon*, Jeffrey Gu*, Ambri Ma, Serena Yeung-Levy, Vincent Sitzmann, Ellen D. Zhong
Oral presentation. paper

NeurIPS 2024 — **CryoBench: Diverse and challenging datasets for the heterogeneity problem in cryo-EM**
Minkyu Jeon, Rishwanth Raghu, Miro Astore, Geoffrey Woollard, Ryan Feathers, Alkin Kaz, Sonya M. Hanson, Pilar Cossio, Ellen D. Zhong
Spotlight. paper / project

NeurIPS 2022 — **SageMix: Saliency-Guided Mixup for Point Clouds**
Sanghyeok Lee*, **Minkyu Jeon***, Injae Kim, Yunyang Xiong, Hyunwoo J. Kim
paper / code

ECCV 2022 — **k-SALSA: k-anonymous synthetic averaging of retinal images via local style alignment**
Minkyu Jeon, Hyeonjin Park, Hyunwoo J. Kim, Michael Morley, Hyunghoon Cho
paper / code

JMIR 2024 — **A multivariable prediction model for mild cognitive impairment and dementia: algorithm development and validation**
Sarah Soyeon Oh, Bada Kang, Dahye Hong, Jennifer Ivy Kim, Hyewon Jeong, Jinyeop Song, **Minkyu Jeon**

Information Sciences 2023 — **Randomly shuffled convolution for self-supervised representation learning**
Youngjin Oh*, **Minkyu Jeon***, Dohwan Ko, Hyunwoo J. Kim
paper / code

IEEE Access 2021 — **Learning to Balance Local Losses via Meta-Learning**
Seungdong Yoa, **Minkyu Jeon**, Youngjin Oh, Hyunwoo J. Kim
paper

In submission — **CHIMERA: Adaptive Cache Injection and Semantic Anchor Prompting for Zero-shot Image Morphing with Morphing-oriented Metrics**
Dahyeon Kye, Jaehun Sung, **Minkyu Jeon**, Jihyong Oh
paper / project

arXiv 2025 — **R3eVision: A Survey on Robust Rendering, Restoration, and Enhancement for 3D Low-Level Vision**
Weeyoung Kwon, Jeahun Sung, **Minkyu Jeon**, Chanho Eom, Jihyong Oh
paper

Research Experience

Prescient Design, Genentech — *Nov 2025–Mar 2026*
Contractor — New York, NY
- Worked on protein representation learning and generative modeling for protein design, with emphasis on modern 3D generative models including diffusion and flow matching.
- Studied limitations of voxel- and structure-space generative models and developed design/evaluation workflows for improved generation performance.

Prescient Design, Genentech — *Jun 2025–Aug 2025*
Research Intern — New York, NY
- Led a project on understanding and improving modern generative models in 3D spaces, with applications to protein generation and design.

AI4ALL — *Jul 2024–Aug 2024*
Instructor — Princeton, NJ
- Instructed high school students in AI and mentored team projects on medical image analysis using deep learning.

Broad Institute of MIT and Harvard — *Jan 2023–Aug 2023*
Associate Computational Biologist; advisor: Dr. Hyunghoon Cho — Cambridge, MA
- Led a project on synthesizing privacy-preserving retinal image datasets with diffusion-based generative models.

Broad Institute of MIT and Harvard — *Sep 2021–Apr 2022*
Visiting Graduate Student; advisor: Dr. Hyunghoon Cho — Cambridge, MA
- Led GAN and GAN-inversion based retinal image synthesis research for privacy-preserving data generation, resulting in an ECCV 2022 publication.

Korea University — *Jan 2020–Aug 2020*
Research Intern; advisor: Prof. Hyunwoo J. Kim — Seoul, Korea
- Conducted meta-learning research for layer-wise optimization and co-authored an IEEE Access publication.
- Designed algorithms, implemented experiments, and contributed to related work, method, and experiment sections.

MakinaRocks — *Mar 2019–Aug 2019*
ML Research Engineer Intern — Seoul, Korea
- Conducted anomaly detection research using variational autoencoders and latent-space regularization.

Korea Institute of Science and Technology — *Jul 2018–Dec 2018*
ML Research Intern; advisor: Dr. Yong Moo Kwon — Seoul, Korea

- Worked on an active chatbot project using rule-based NLP and deep learning methods for situation-aware conversation.

Awards and Service

2024–present	**Asan Foundation Biomedical Science Scholarship**. Three-year doctoral scholarship.
2023–present	**Reviewer**. CVPR (2026), ECCV (2026), ICLR (2025, 2026), NeurIPS (2025, 2026), NeurIPS Workshop (2023, 2024), Information Sciences (2024).
Aug 2024	**Invited talk**. Flatiron Institute, Simons Foundation.
Feb 2024	**Invited talk**. CMU MetaMobility Lab.
Sep 2019	**Dean's List**. Department of Applied Statistics, Konkuk University.
Sep 2019	**Konkuk University Scholarship**. 40% tuition scholarship.
Jul 2016	**Certificate of Commendation**. Republic of Korea Marine Corps.

Teaching

Fall 2024	Teaching Assistant, COS302: Mathematics for Numerical Computing and Machine Learning, Princeton University.
Spring 2021	Teaching Assistant, COSE361: Artificial Intelligence, Korea University.
Fall 2020	Teaching Assistant, AAA712: AI Security, Korea University.
Fall 2020	Tutorial Teaching Fellow, Deep Learning, Korea University.

Skills

Programming	Python, PyTorch, TensorFlow, Keras, R, C, Git.
Systems	Linux, macOS, Windows, scientific computing workflows.
Research	Generative modeling, diffusion/flow matching, protein design, cryo-EM reconstruction, representation learning, 3D vision.

Additional Experience

I-Corps Program *May 2021–Aug 2021*

Received $35,000 in government support for customer discovery and technology-based startup exploration.

Republic of Korea Marine Corps *Mar 2015–Dec 2016*

Military service, Yeonpyeong Island, South Korea; received commendation for communication and performance during early warning situations.